ENGINEERING MINI-PROJECT REPORT

Battery Health Prediction Using Machine Learning

Regression-based battery capacity estimation and health assessment

Submitted by

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Executive Summary

Battery systems gradually lose usable capacity as charge-discharge cycling continues. Because capacity is closely associated with the practical condition of a battery, estimating it from operating measurements can provide a useful data-driven indicator of battery degradation.

This project develops a supervised machine-learning regression workflow for predicting battery capacity from cycle and operating parameters. The repository contains a 10,000-record dataset covering 40 battery identifiers and cycles from 1 to 250. The modeling workflow uses cycle number, average voltage, average current, temperature, charge time, discharge time, and internal resistance as predictors, while battery capacity in ampere-hours is the target.

The implementation performs exploratory data analysis, mean-based missing-value treatment, removal of the battery identifier from the predictive feature set, chronological train-test separation, feature scaling for Linear Regression, and comparison with a Random Forest Regressor. The notebook reports MAE, RMSE, and R^2 for both models. Linear Regression achieves the better reported MAE and R^2 under the implemented evaluation.

> **Important implementation note:** the notebook's Random Forest evaluation prints the RMSE using the Linear Regression prediction variable. Therefore, the reported Random Forest RMSE should not be interpreted as an independent Random Forest RMSE without correcting that line and rerunning the notebook.

1. Introduction

Lithium-ion and other rechargeable batteries are widely used in portable electronics, electric mobility, energy storage, and embedded systems. Their electrical characteristics change with repeated cycling, operating conditions, and accumulated usage. A battery that has lost a significant portion of its original capacity may provide shorter operating time and require earlier replacement or maintenance.

Conventional battery-health assessment can require repeated measurements and detailed electrochemical analysis. A machine-learning approach offers a complementary data-driven method: historical operating measurements can be mapped to a measurable battery-performance quantity such as capacity.

The present project focuses on capacity prediction rather than claiming a complete battery-management system. The predicted capacity can subsequently be used to derive or interpret a battery health indicator, while the dataset also contains a recorded SOH percentage for analysis.

2. Problem Definition

The engineering problem is to estimate battery capacity from measurable operating and performance parameters collected across charge-discharge cycles. The intended system accepts numerical battery measurements and produces an estimated capacity in ampere-hours.

The problem is formulated as regression because capacity_Ah is a continuous numerical quantity. The objective is therefore not simply to label a battery as healthy or unhealthy, but to estimate the continuous capacity value associated with its operating state.

2.1 Project Domain

Battery analytics, energy systems, battery-management applications, and applied machine learning.

2.2 Objectives

- Understand the structure and quality of the battery-health dataset.
- Inspect missing values, descriptive statistics, distributions, and relationships among battery parameters.
- Prepare the data for supervised regression without using the battery identifier as a predictive variable.
- Use battery cycle information and measurable electrical/thermal parameters as model inputs.
- Compare Linear Regression and Random Forest Regression.
- Evaluate predictions using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2.
- Interpret the model results in the context of battery degradation and capacity estimation.
- Identify limitations and practical improvements for a more robust battery-health solution.

2.3 Expected Inputs

- Cycle number
- Average voltage (V)
- Average current (A)
- Temperature (°C)
- Charge time (hours)
- Discharge time (hours)
- Internal resistance (Ω)

2.4 Expected Output

Predicted battery capacity in ampere-hours (Ah).

2.5 Engineering Decision Supported

The predicted capacity can be used as an application-level indicator of battery condition. A lower capacity estimate relative to a battery's original or reference capacity indicates greater degradation and can support inspection, maintenance, replacement planning, or further battery-health analysis.

3. Dataset Description

The repository includes battery_health_data.csv. The dataset contains 10,000 records and 10 original columns. Battery identifiers range from 1 to 40, and cycle numbers range from 1 to 250. Several sensor-related columns contain missing observations, while the target capacity_Ah is complete.

Dataset Property	Value
File	battery_health_data.csv

Records	10,000
Original columns	10
Battery IDs	1–40
Cycle range	1–250
Target	capacity_Ah
Additional health indicator	soh_percent
Missing data	Present in six sensor-related columns

3.1 Dataset Variables

Feature	Description	Role
battery_id	Unique battery identifier.	Identifier; removed before modeling
cycle	Charge/discharge cycle number.	Predictor
avg_voltage_V	Average battery voltage in volts.	Predictor
avg_current_A	Average current in amperes.	Predictor
temperature_C	Battery temperature in degrees Celsius.	Predictor
charge_time_hr	Charging duration in hours.	Predictor
discharge_time_hr	Discharging duration in hours.	Predictor
internal_resistance_ohm	Internal resistance in ohms.	Predictor
capacity_Ah	Battery capacity in ampere-hours.	Target
soh_percent	State-of-health percentage recorded in the dataset.	Context/analysis

4. Data Exploration

The notebook performs systematic inspection before model development. This includes viewing the first and last records, checking dataset dimensions and data types, computing descriptive statistics, counting missing values, inspecting distributions, examining battery-cycle behavior, and visualizing relationships between variables.

4.1 Exploratory Analysis Performed

- Dataset shape and structure inspection.
- Statistical summary of numerical variables.
- Missing-value analysis by column.
- Battery-cycle and degradation trend analysis.
- Battery capacity distribution analysis.
- Visual comparison of operating parameters.

- Correlation and relationship analysis.
- Actual-versus-predicted comparison after model evaluation.

4.2 Key Dataset Observations

Variable	Mean	Minimum	Maximum
Cycle	125.50	1.00	250.00
Average voltage (V)	4.1238	3.9881	4.2500
Average current (A)	1.5183	1.0520	1.9203
Temperature (°C)	25.5292	20.7060	29.4290
Charge time (hr)	2.3018	1.8948	2.7467
Discharge time (hr)	1.2717	0.8831	1.6599
Internal resistance (Ω)	0.0600	0.0410	0.0811
Capacity (Ah)	1.7398	1.3287	2.1419
SOH (%)	86.9867	66.66	100.70

> **Interpretation:** the dataset spans a substantial range of battery condition. Capacity decreases from high values at early cycles toward lower values at later cycles, making cycle number an important degradation-related input.

5. Data Preprocessing

Preprocessing is required because several sensor columns contain missing observations and because the battery_id field identifies a battery rather than representing a physical measurement that should be learned as a numeric feature.

5.1 Missing-Value Treatment

Column	Missing values	Treatment
avg_voltage_V	200	Filled with column mean
avg_current_A	250	Filled with column mean
temperature_C	300	Filled with column mean
charge_time_hr	200	Filled with column mean
discharge_time_hr	250	Filled with column mean
internal_resistance_ohm	300	Filled with column mean
capacity_Ah	0	No imputation required
soh_percent	0	No imputation required

The notebook applies mean imputation to the six sensor-related columns. After preprocessing, the dataset contains no missing values.

5.2 Identifier Removal

The battery_id column is removed before model development. Treating an identifier as a numeric feature could allow the model to learn arbitrary differences between IDs rather than meaningful physical behavior. The resulting modeling dataset contains nine columns, including the target.

5.3 Train-Test Separation

Instead of a random split, the notebook creates a cycle-based chronological split. The maximum cycle is 250, and 80% of this range corresponds to cycle 200. Records through cycle 200 form the training set, while cycles 201–250 form the test set. This produces 8,000 training records and 2,000 test records.

> **Why this matters:** a cycle-based split better reflects a degradation setting in which earlier battery cycles are used to learn patterns and later cycles are treated as unseen observations.

6. Feature Engineering and Feature Selection

The project intentionally uses a compact, physically interpretable feature set rather than generating a large number of derived variables. The final predictors are the cycle number and six operating or electrical measurements.

Predictor	Reason for inclusion
cycle	Captures progression through repeated battery use and degradation.
avg_voltage_V	Represents the average electrical potential during the recorded operating cycle.
avg_current_A	Captures electrical loading/current behavior.
temperature_C	Represents thermal operating conditions.
charge_time_hr	Provides information about charging behavior.
discharge_time_hr	Provides information about discharge behavior.
internal_resistance_ohm	Captures an electrical characteristic associated with battery condition.

The dataset's soh_percent field is not used as a predictor in the capacity model. It is retained as a contextual battery-health variable because the project's primary target is capacity_Ah.

7. Machine Learning Technique

Regression is the appropriate primary formulation because the target variable, capacity_Ah, is continuous. The project compares a simple linear baseline with a nonlinear tree-based ensemble.

7.1 Candidate Algorithms

Model	Purpose	Strength	Limitation
Linear Regression	Baseline continuous	Simple, fast, interpretable.	May not capture complex

	prediction.		nonlinear relationships.
Random Forest Regressor	Nonlinear ensemble comparison.	Captures interactions and nonlinear patterns.	More complex and sensitive to evaluation design; less directly interpretable.

7.2 Linear Regression Pipeline

StandardScaler → Linear Regression

The predictor variables are standardized before Linear Regression. The scaler is fitted on the training data and then applied to the test data, maintaining a consistent transformation.

7.3 Random Forest Configuration

RandomForestRegressor(n_estimators=100, random_state=42)

The Random Forest model is trained on the same seven predictors and the same chronological training/test partition. The use of 100 trees provides an ensemble-based nonlinear model for comparison.

8. Implementation

The project is implemented in Python using Jupyter Notebook. The repository contains the notebook, dataset, and README documentation.

Component	Technology / File
Programming language	Python
Notebook	main.ipynb
Dataset	battery_health_data.csv
Data handling	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine learning	Scikit-learn
Repository	GitHub – Battery_Health_Predictor

8.1 Processing Workflow

1. Load battery-health dataset.
2. Inspect dimensions, data types, summary statistics, and missing values.
3. Fill missing sensor values using column means.
4. Remove battery_id from the predictive dataset.
5. Define seven predictors and capacity_Ah as the target.
6. Split the data chronologically using cycle 200 as the boundary.
7. Standardize predictors for Linear Regression.
8. Train Linear Regression and Random Forest models.

9. Generate test-set predictions.
10. Calculate MAE, RMSE, and R^2.
11. Compare model behavior and interpret the engineering meaning.

9. Results

The repository README and executed notebook provide numerical evaluation results. The test set contains 2,000 observations from cycles 201–250.

Model	MAE (Ah)	RMSE (Ah)	R^2
Linear Regression	0.073357	0.086489	0.273054
Random Forest	0.084841	0.086489*	0.008143

* The notebook's Random Forest RMSE output is produced using the variable y_pred, which contains Linear Regression predictions. It therefore duplicates the Linear Regression RMSE and should be recalculated using rf_pred before being treated as a valid Random Forest RMSE.

9.1 Result Interpretation

- Linear Regression has the lower reported MAE (0.073357 Ah), meaning its average absolute error is smaller in the implemented test evaluation.
- Linear Regression also has the higher reported R^2 (0.273054), indicating that it explains more of the observed capacity variation than the tested Random Forest under this implementation.
- The Random Forest R^2 value of 0.008143 is very low for this evaluation, so its current configuration does not provide strong explanatory performance on the test set.
- The identical RMSE values should not be used to claim equal Random Forest and Linear Regression performance because the notebook contains an evaluation-variable error.

10. Evaluation

10.1 Evaluation Metrics

Metric	Meaning in this project
MAE	Average absolute difference between predicted and observed battery capacity, expressed in Ah.
RMSE	Penalizes larger prediction errors more strongly and expresses the result in Ah.
R^2	Measures how much of the target variation is explained relative to a baseline.

10.2 Evaluation Strategy

The evaluation is performed on later cycles rather than randomly selected rows. This is a useful design choice for degradation-related data because it approximates predicting later battery behavior from earlier cycles. However, the dataset contains multiple battery IDs, so future work should additionally evaluate generalization to completely unseen batteries.

11. Engineering Interpretation

The practical value of the project is not limited to a numerical prediction. Battery capacity is a direct indicator of how much charge the battery can store relative to its usable condition. As a battery ages, capacity generally decreases, and this reduction can affect runtime and application reliability.

- A predicted capacity close to the battery's original capacity indicates comparatively better retained performance.
- A substantially reduced capacity can be used as a trigger for inspection, maintenance, or replacement planning.
- Cycle number provides a direct measure of accumulated use and is therefore relevant to degradation modeling.
- Internal resistance, temperature, voltage, current, and charge/discharge durations provide additional context for the model.
- The model should be treated as a decision-support tool rather than as a replacement for laboratory battery characterization or a certified BMS.

11.1 Relationship to State of Health

The dataset also contains soh_percent. A common capacity-based interpretation of state of health is to compare current usable capacity with a reference or initial capacity. The repository notebook demonstrates a capacity-based SOH calculation using predicted capacity and an initial-capacity reference. This should be treated as an analytical estimate and validated carefully before operational use.

12. Limitations

- The current model predicts capacity rather than directly forecasting future SOH or Remaining Useful Life (RUL).
- The dataset is limited to the operating conditions represented in the supplied records; performance may change under different cells, temperatures, loads, or chemistries.
- Mean imputation can hide systematic missingness and may reduce the information contained in sensor gaps.
- The battery_id is removed, but the chronological split is not a battery-wise holdout. The reported performance therefore does not establish generalization to completely unseen batteries.
- The current Random Forest evaluation contains an RMSE implementation error that should be corrected before using the comparison as a final benchmark.
- The R^2 values indicate that the current feature/model setup leaves substantial variation in capacity unexplained, particularly for the Random Forest implementation.
- The project does not include uncertainty estimates, confidence intervals, or explicit failure detection for out-of-distribution inputs.
- Predictions should not be treated as a certified safety measurement or as a replacement for electrical protection and battery-management hardware.

13. Recommendations and Future Improvements

- Correct the Random Forest RMSE calculation by evaluating rf_pred against y_test.
- Perform systematic hyperparameter tuning for Random Forest and compare additional regressors such as Gradient Boosting, Extra Trees, or XGBoost where permitted.
- Use cross-validation that respects battery identity and temporal order.
- Introduce battery-wise holdout testing so that test batteries are entirely unseen during training.
- Engineer degradation-aware features such as rolling capacity trends, capacity fade rate, cycle-normalized resistance changes, and temperature statistics.
- Evaluate direct SOH prediction and Remaining Useful Life estimation as separate modeling tasks.
- Add model explainability methods to understand how voltage, current, temperature, timing, resistance, and cycle number influence predictions.
- Build a web dashboard or Streamlit application for interactive battery-health monitoring.
- Add input validation to detect physically impossible or clearly corrupted measurements.
- Monitor model drift and periodically retrain when new battery-cycle data become available.
- Compare model predictions with laboratory or BMS measurements before operational deployment.

14. Conclusion

This project demonstrates an end-to-end machine-learning workflow for battery capacity prediction. The work begins with a structured battery-cycle dataset, examines data quality and missing values, removes the battery identifier from the modeling feature set, and selects seven physically meaningful predictors.

The target is treated as a continuous quantity, making regression the appropriate primary formulation. Linear Regression and Random Forest Regression are implemented and evaluated on a chronological test set containing later battery cycles. Under the reported implementation, Linear Regression performs better than Random Forest in both MAE and R^2. The reported Random Forest RMSE requires correction because of a notebook variable-reference issue.

Overall, the project provides a useful foundation for battery-health analytics. Its strongest next step is to move from a basic capacity-prediction experiment toward battery-wise validation, degradation-aware feature engineering, robust model comparison, uncertainty analysis, and an interactive monitoring application.

References

1. Omkar Satpute, Battery_Health_Predictor, GitHub repository: https://github.com/omkarsatpute18/Battery_Health_Predictor

2. Battery Health Predictor project dataset: battery_health_data.csv, contained in the project repository.

3. Project notebook: main.ipynb, contained in the Battery_Health_Predictor repository.

4. Pune Vidyarthi Griha's College of Engineering, Technology and Management, Department of Electronics and Telecommunication Engineering, mini-project report guideline structure supplied by the student.

Appendix A – Requirement Compliance Checklist

Requirement / Report Element	Status	Evidence in this report
Engineering problem defined	Completed	Section 2
Objectives stated	Completed	Section 2.2
Dataset documented	Completed	Section 3
Features explained	Completed	Section 3.1
Data quality / missing values	Completed	Sections 4 and 5
Exploratory analysis	Completed	Section 4
Preprocessing decisions	Completed	Section 5
Feature selection	Completed	Section 6
ML technique justified	Completed	Section 7
Model implementation	Completed	Section 8
Evaluation metrics	Completed	Section 9 and 10
Engineering interpretation	Completed	Section 11
Limitations	Completed	Section 12
Recommendations	Completed	Section 13
Conclusion	Completed	Section 14
Submission checklist	Included	Appendix C

Appendix C – Final Submission Checklist

- Attach main.ipynb with executed outputs.
- Attach battery_health_data.csv or provide the repository/dataset source as required by the submission process.
- Verify that all numerical metrics in the report match the executed notebook.
- Correct and rerun the Random Forest RMSE calculation before final submission.
- Insert selected notebook visualizations if the college requires printed figures.
- Ensure the final report, notebook, dataset, and repository README describe the same workflow.
- If a future Streamlit dashboard is added, include the application file and dependency list with the submission.

> **Source note:** The structure and section hierarchy of this report were adapted from the supplied sample report, while the technical content was rewritten for the Battery Health Predictor project and checked against its GitHub README, dataset, and executed notebook. The original sample's visual styling, font choices, and color scheme were not copied.